

Lists

Understand Lists (and arrays)

Declare and initialize Lists

Use Lists in a program

Storing related data

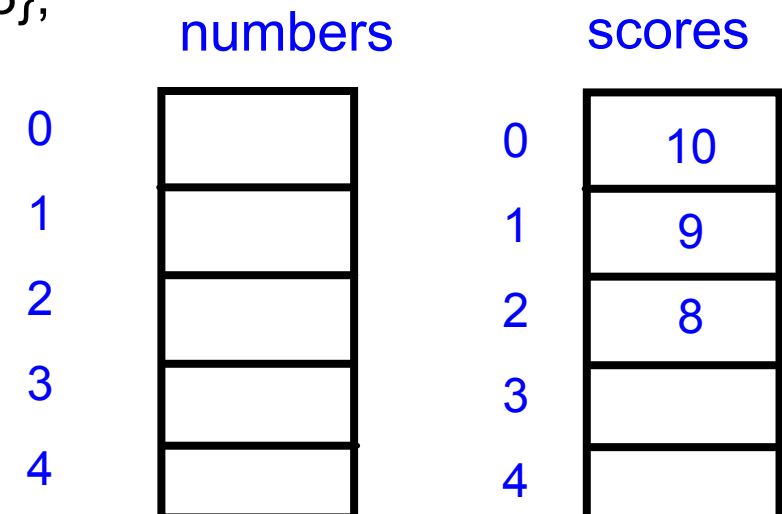
Lists (and arrays) are used to hold related data. We use the same name for data and an index is used to access each different piece of data.

```
int score01, score02, score03;
```

```
List<int> numbers = new List<int>();
```

```
List<int> scores = new List<int>{10, 9, 8};
```

```
Console.WriteLine(scores[2]);
```



Arrays

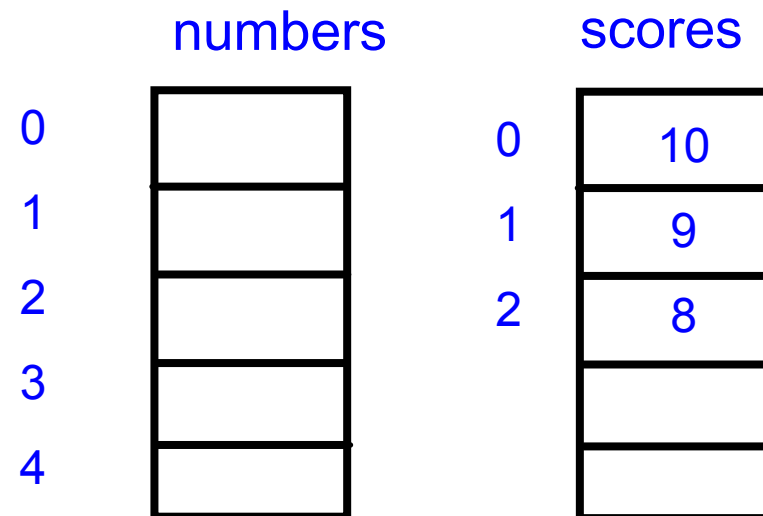
Arrays are a different way to store related data. The biggest difference between arrays and Lists is that arrays are not dynamic and Lists are.

```
int[] array = new int[];    // does not work
```

```
int[] numbers = new int[5]; // array of 5 ints
```

```
int[] scores = {10, 9, 8};
```

```
Console.WriteLine(scores[2]);
```



Lists - declaring and initializing

```
List<int> scores = new List<int>(); // declare
```

```
List<double> nums = new List<double>{1.1,2.2};
```

```
List<bool> TFs = new List<bool>{true,true,false};
```

etc.

```
List<int> list;
```

...

```
list = new List<int>();
```

	numbers	scores
0		10
1		9
2		8
3		
4		

Lists

```
List<int> scores = new List<int>();
```

```
scores.Add(9);
```

```
scores.Add(8);
```

```
int index = 0;
```

```
scores.Add(index, 10);
```

```
scores.Add(5);
```

```
scores.Add(4);
```

```
scores.Add(2);
```

```
scores.Remove(5);
```

```
scores.RemoveAt(2);
```

↖ value

↑ at index

numbers

0	
1	
2	
3	
4	

scores

0	10
1	9
2	8 4
3	5 2
4	4

Lists - indexing, output

```
List<int> scores = new List<int>{10,9,8,7,6};
```

```
for(int index = 0; index < scores.Count; index++)
```

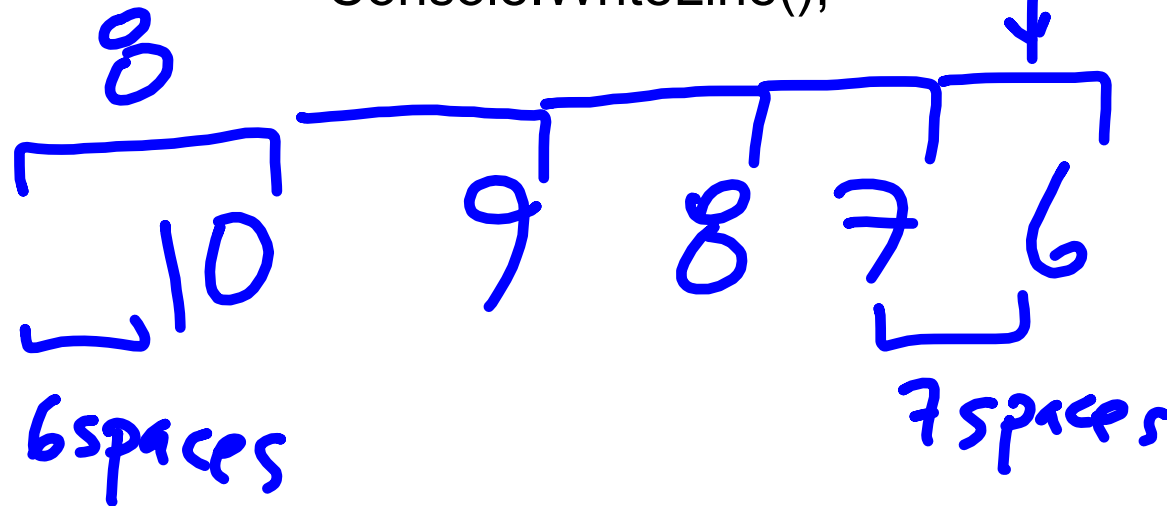
```
{
```

```
    Console.WriteLine($"{list[index],8}");
```

```
}
```

```
Console.WriteLine();
```

how many elements in List



numbers	scores
0	10
1	9
2	8
3	7
4	6

Lists - indexing, output

```
List<double> ds = new List<double>{10.0,9,8,7,6};
```

```
for(int index = 0; index < scores.Count; index++)
```

```
{
```

```
    Console.Write($"{list[index],8:0.00}");
```

```
}
```

```
Console.WriteLine();
```

Handwritten output showing the formatted values of the list elements:

10.00 9.00 8.00

Below the first value (10.00) is a bracket with the number 3. Below the second value (9.00) is a bracket with the number 4. Below the third value (8.00) is a bracket with the number 4. Above the third value (8.00) is a bracket with the number 8.

numbers

0
1
2
3
4

ds

0
1
2
3
4

10
9
8

Lists versus Arrays

```
List<int> scores = new List<int>{10,9,8,7,6};
```

```
int [] numbers = {10,9,8,7,6};
```

What is the difference?

numbers[0] = 30;
X number[5] = 3;

	numbers		scores
0	1020	0	10
1	9	1	9
2	8	2	8
3	7	3	7
4	6	4	6